

High-intensity focused ultrasound (HIFU) treatment for early prostate cancer

High intensity focused ultrasound (HIFU) is a treatment that is sometimes used to treat early prostate cancer or prostate cancer that has come back.

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What is high-intensity focused ultrasound (HIFU) treatment?

High-intensity focused ultrasound (HIFU) treatment is sometimes used to treat:

- [early prostate cancer](#)
- [prostate cancer that has come back](#) after radiotherapy or brachytherapy.

This treatment is done by a urologist or urology surgeon who specialises in treating prostate cancer.

HIFU not routinely available in all NHS hospitals. It is important to talk to your urologist and specialist nurse first to find out whether a treatment is available in your area. Usually, you can only have HIFU on the NHS as part of a [clinical trial](#). Some private clinics also offer these treatments.

This treatment is not suitable for everyone. It depends on where the cancer is in the prostate, and the [risk group](#) of the cancer. Your doctor can explain more about this.

When is HIFU treatment used?

HIFU can be used when there is only 1 small area of cancer in the prostate. This is called focal therapy. It can sometimes be used to treat the whole prostate when the cancer is in more than 1 area.

If there are areas of cancer that are growing quickly ([high grade](#)) you can have focal treatment to these areas. Very slow growing ([low grade](#)) areas of cancer may not be treated. You may have fewer side effects because less healthy tissue is damaged.

Before treatment

The urologist will explain what will happen and tell you about the possible side effects and risks. You will be given written information to take away. It is important you have all the information you need to decide about having treatment. We have more information about talking to your healthcare team and [asking questions](#). We also have more information to help if you need support in [finding and understanding information](#).

You may also have a pre-operative assessment. This may involve having some tests to check your general health and fitness. You can also find out more about how to prepare for your treatment at this appointment.

You may be encouraged to start [pelvic floor exercises](#) to strengthen the pelvic floor muscles. You can do these at home. This can help reduce [urinary leakage \(incontinence\)](#) after treatment.

How is HIFU treatment given?

You usually have HIFU under a general anaesthetic, which means you are not awake when you have the procedure. Some people have a spinal anaesthetic – this is an injection into the spine. This makes you numb from the waist down. With a spinal anaesthetic you are often given some sedation to make you relaxed and sleepy, so you are less aware of what is happening. You may be awake but you will not feel any pain. You can usually go home when you have recovered from the anaesthetic.

The surgeon passes a probe into the anus to the lower bowel (rectum). It produces beams of high-energy ultrasound that heat and destroy the area of prostate cancer cells. The probe is surrounded by a cooling balloon so the high-energy beams are only given to the areas with cancer. This means the normal prostate tissue is protected.

Side effects of HIFU treatment for prostate cancer

Urinary problems

During surgery, you will have a tube put in to drain urine from your bladder. This is called a catheter. The catheter keeps your bladder empty while the area heals. You might notice a small amount of blood or debris in your pee. Try to drink plenty of fluids to help keep the catheter draining well.

The nurses on the ward will show you how to look after your catheter before you go home. They will give you spare bags to take home.

The nurses will give you details of who to contact if you have any problems or concerns about your catheter. The catheter is usually removed 1 to 2 weeks after surgery. The nurses may give you a clinic appointment date to have this done. Ask them before you go home or call the number you have been given if you are not sure about anything to do with your catheter.

You may leak urine for a while after the catheter is removed. You may need to wear a pad in your underwear while this gets better. Your surgeon and specialist nurse can talk to you about what to expect.

Sometimes a [transurethral resection of the prostate \(TURP\)](#) operation is done before HIFU to reduce bladder problems afterwards.

Urinary retention



This happens when you cannot empty your bladder properly after the catheter is removed. HIFU can cause the prostate to swell. This may block the tube from the bladder that urine passes through ([urethra](#)). If your urine flow is weak or very slow, contact the number you were given for advice. If you cannot pass urine at all, contact your doctor or nurse straight away or go to your nearest A&E (emergency department).

Long-term urinary problems



Some people may have long term [urinary problems](#) such as leakage (incontinence) after HIFU. This is more likely if you have already had external beam radiotherapy. Another late effect of HIFU can be a narrowing in the urethra. This is called a urethral stricture. If you are having problems passing urine, talk to your GP, urologist or clinical nurse specialist.

Pain and bleeding

After treatment, you may have some pain in your lower tummy or lower bowel (rectum). The surgeon or nurse can advise you about which painkillers you can take. You may have some bleeding from the anus, but this is not common.

Erection problems

HIFU can cause problems getting an erection. Your surgeon and clinical nurse specialist can talk to you about this. Different [treatments](#) can help if you have erectile dysfunction (ED).

Ejaculation

After HIFU, there is a small chance of having little or no ejaculation. This is called a dry ejaculation or dry orgasm. It may cause some discomfort at first, but this usually improves with time. You may pass a small amount of urine when you orgasm. Talk to your urologist or specialist nurse if you are having this problem. They may be able to give you some advice.

A dry ejaculation can affect fertility. This is the ability to make someone pregnant. It is important to talk to your urologist or specialist nurse about fertility before treatment. It may be possible to store sperm before treatment. We have more information about [fertility](#).

Rectal fistula

Rarely, HIFU can damage tissues surrounding the prostate and cause a hole between the rectum and the urethra. This is called a rectal fistula. If you have a rectal fistula, you will need an operation to repair it.

Benefits and disadvantages of HIFU

If you are thinking about having HIFU, your cancer team can explain the possible benefits and disadvantages.

They will also tell you about the other treatment options available to you. Doctors do not know how effective HIFU is when compared to existing treatments for early prostate cancer.

Side effects, such as [erection difficulties](#) and [urinary problems](#) may be less of a problem with HIFU. But they may still affect you. If you have already had other treatment, such as radiotherapy or brachytherapy, the side effects may be worse.

Some people may prefer to have HIFU even though there is not enough evidence about them yet. They may feel there is less risk of certain side effects and want to avoid surgery or radiotherapy. Or it may be because these treatments can be given in a short hospital stay and recovery is usually quick.

If the cancer comes back after HIFU, you may still be able to have treatment with [surgery](#) or [radiotherapy](#). You will have your PSA level checked regularly after treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

National Institute for Health and Care Excellence (NICE). Prostate cancer: diagnosis and management. NICE Guideline [NG131]. Published: 09 May 2019. Last updated: 15 December 2021. Available from: www.nice.org.uk/guidance/ng131 [accessed March 2024].

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[Cancer Information Development team](#)

Author

[Dr Ursula McGovern](#)

Reviewer

Consultant Medical Oncologist & Honorary Associate Professor
University College Hospitals, London

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences

- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help